

Lab 11: Inference for Regression in R

STAT 141, Week 12

Your name here

In this lab, you will

1. Recall how to build and interpret linear regression models
2. Do inference for regression with theory-based methods

Add any collaborators here!

I collaborated with...

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

```
# Load the necessary packages
library(tidyverse)
library(knitr)
library(moderndiver)
library(infer)
library(openintro)
```

Exercises

Background: Grading the Professor

Many college courses conclude by giving students the opportunity to evaluate the course and the instructor anonymously. The use of these student evaluations as an indicator of course

quality and teaching effectiveness is often criticized because these measures may reflect the influence of non-teaching related characteristics. The article titled, “[Beauty in the classroom: instructors’ pulchritude and putative pedagogical productivity](#)” by Hamermesh and Parker found that instructors who are viewed to be better looking receive higher instructional ratings. **In this lab we will analyze the data from this study in order to learn what goes into a positive professor evaluation.**

The data were gathered from end of semester student evaluations for a large sample of professors from the University of Texas at Austin. In addition, six students rated the professors’ physical appearance. The result is a data frame where each row contains a different course and columns represent variables about the courses and professors. Let’s load the data:

```
data(evals)
```

We have observations on 21 different variables, some categorical and some numerical. The meaning of each variable can be found by bringing up the help file (select the help file in the `moderndive` package):

```
?evals
```

Exercise 1: Inference for simple linear regression of score vs beauty

a. We will start by investigating the data with simple linear regression. First, use `ggplot` to make a scatterplot of `score` (our response variable) by `btv_avg` (our explanatory variable) and *display a regression line*. [Code only for this question]

b. Use `lm()` to create a regression model predicting `score` as a function of `btv_avg`, and display the estimated coefficients using the `get_regression_table()` function. [Code only for this question]

c. Write down the equation for the **fitted** linear model, being sure to include the estimated coefficients in the equation. [Text only for this question]

d. Interpret the (i) intercept and the (ii) slope on `btv_avg`, *in context*. Remember to be precise about your language, and revisit the linear regression slides for a refresher if needed (ex. slides 17-18 on 2/23). [Text only for this question]

e. Formally assess whether `btv_avg` is a meaningful predictor of `score` by stating a set of two hypotheses, writing down a *theory-based* p-value (your previous code should have given it to you!), and interpreting that p-value. [Text only for this question]

f. State and interpret the *theory-based* 95% confidence interval for the `btv_avg` coefficient. **How does this result align with your conclusion from the previous part?** [Text only for this question]

g. Is `btv_avg` a “practically significant” predictor of `score`? In other words, does `btv_avg` have a “big” effect on `score`? [Text only for this question]
